

**CTSpinoPelvic1K: a fused spine and pelvis CT segmentation dataset built for
lumbosacral transitional anatomy**

[Author names removed for double-anonymised review]

[Affiliations removed for double-anonymised review]

(Dated: 2 September 2026)

Purpose: Lumbar levels cannot be numbered reliably when a lumbosacral transitional vertebra (LSTV) is present. Sacralisation and lumbarisation are different developmental events and leave different traces, but those traces are not decisive in every case, and the count that would arbitrate is taken downward from C2 — the one landmark present in every spine and never duplicated¹. A field of view that begins in the thorax contains no such landmark, so the name rests on morphology alone exactly where morphology is least certain. Public collections either omit the pelvis (CTSpine1K², VerSe³) or omit the classes a transitional vertebra requires: TotalSegmentator⁴, the largest whole-body collection, provides no sixth lumbar vertebra and no class for a rib arising from a lumbar level, and VerSe excludes vertebrae fused to the sacrum by design⁵. CTSpinoPelvic1K provides 802 CT records carrying radiologist-sourced spine *and* pelvic annotation on one coordinate frame, with per-level ribs and femora. Four classes exist so that a variant need not be forced into an ordinary level: a sixth lumbar vertebra, a thirteenth thoracic vertebra, the first sacral segment held separate from the sacrum, and left and right lumbar ribs. The primary use is the lumbosacral interface — sacral morphology measured against the lumbar spine, for pelvic fixation and lumbar surgical planning — and the transitional labelling is what makes that interface trustworthy, since a trajectory cannot be planned across a junction whose segments cannot be named. Because every record follows one screening protocol, between-scan variation reflects anatomy rather than acquisition, which suits the collection to learning morphology directly.

Acquisition and Validation Methods: Records were built by patient-level crosswalk of three public collections — TCIA CT COLONOGRAPHY imaging, CTSpine1K² vertebral labels, and CTPelvic1K⁶ pelvic labels — with labels placed on the acquisition carrying the highest bone coverage and unified under a VerSe-native scheme. Validation is threefold: per-case geometric invariants (802/802 pass); rib-vertebra incidence across 5,749 evaluable ribs (2 residual offsets, 0.035%); and agreement of derived spinopelvic measures with published reference values.

Data Format and Usage Notes: Gzipped NIfTI image/label pairs sharing an identical affine, canonicalised to PIR, with frozen patient-grouped LSTV-stratified five-fold cross-validation splits and a reference loader. Archived at 10.5281/zenodo.22139642.

Potential Applications: Segmentation and level-numbering benchmarks, anatomical variant studies, surgical planning, and opportunistic screening research. *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; the soft-tissue identifiers are declared but unpopulated; the splits are cross-validation folds and include no held-out test set; the rib layer is a reviewed pseudolabel whose review was triaged by an automated rule rather than exhaustive; and the Castellvi grades are one reader's, with confirmation by a board-certified neuroradiologist in progress.

6 I. INTRODUCTION

7 Labelling a lumbar vertebra requires counting, and counting requires two fixed ends.
8 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, but
9 *enumeration anomalies* — eleven or thirteen thoracic, four or six lumbar — are common, and
10 the transitional phenotypes that produce them sit at the two ends of the lumbar column: the
11 lumbosacral transitional vertebra (LSTV) below and the thoracolumbar transitional vertebra
12 (TLTV) above. In the presence of an LSTV the lower end is not where it is assumed to be.
13 The two phenotypes are distinct developmental events — a lumbar segment assimilating
14 to the sacrum, or a sacral segment separating from it — and they leave different traces:
15 a sacralised L5 tends toward an anteriorly wedged body, a reduced disc beneath it and
16 hypoplastic or absent facet joints, while a lumbarised S1 tends toward a squared body in the
17 sagittal plane, well formed lumbar-type facets and a full-height disc¹. What counting settles
18 is the *name*. The difficulty is that these traces are not decisive in every case, so the name and
19 the morphology can disagree, and the count that would arbitrate is taken downward from
20 C2 — unavailable in any spine-limited acquisition. LSTV is common, reported between 4%
21 and 30% depending on definition, and wrong-level surgery is its most serious consequence.

22 Existing public collections do not make this tractable, and not because they are small
23 (Fig. 2): the spine collections omit the pelvis, the pelvic ones carry no vertebral count,
24 and the whole-body schemes have neither an L6 class nor a class for a rib borne by a
25 lumbar vertebra, so a six-lumbar spine cannot be written down in them at any segmentation
26 quality. VerSe³ is the instructive case: it enriched for anatomical variants deliberately and
27 grades transitional vertebrae by Castellvi, then uses the grade to decide what to omit. Its
28 descriptor⁵ states that vertebrae showing partial fusion with the sacrum, grades III and IV,
29 were not segmented; nor is the sacrum. Those grades are 25 of the 33 graded records here.

30 CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and
31 gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra,
32 and a rib borne by a lumbar vertebra. A scheme without those classes must record such
33 anatomy as something it is not.

34 *Where the material is.* The release described here (v7) is archived at [https://doi.](https://doi.org/10.5281/zenodo.22139642)
35 [org/10.5281/zenodo.22139642](https://doi.org/10.5281/zenodo.22139642), which resolves to the current version; the exact version
36 described by this manuscript is 10.5281/zenodo.22242745. Documentation, the label scheme,

the reference loader and the per-record quality-control tables are distributed with the archive and are described in Sec. III.

I.A. Vertebra labelling when the scan cannot settle the count

The limitation, stated plainly. No scan in this corpus contains C2 or the whole cervical spine, so the conventional count cannot be performed. That is a property of abdominal imaging rather than of the annotation, and it means a variant at the thoracolumbar junction cannot be *resolved by counting* here at all: a thirteenth thoracic vertebra, a lumbar rib, and a *stump rib* — a hypoplastic twelfth rib, one of the cardinal indicators of a thoracolumbar transitional vertebra — produce overlapping appearances.

What the release does instead. No vertebral label is asserted where it cannot be supported. Every quantity is expressed against the two landmarks present in essentially every abdominal scan — the sacrum below and the lowest rib-bearing vertebra above — so the description is positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes according to whether a reader would have said L5 or S1, so cases that disagree about the name remain comparable. Where the junction is in view the count is determined and the conventional name follows, and the two descriptions agree.

And the morphology may be local rather than global. Counting is not the only route to an identity. A thoracic vertebra differs in shape from a lumbar one — in the span of its transverse processes relative to its body, in the proportions of its endplates and canal — and if that difference holds at the boundary itself, the phenotype is decidable from the vertebra in front of the reader without any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is deliberately removed, because a classifier given raw millimetres separates thoracic from lumbar immediately and has learned nothing that helps at the junction, where the two are nearly the same size.

That is reported as a property of the released measurements rather than as a method: the information needed to settle these variants is present locally.

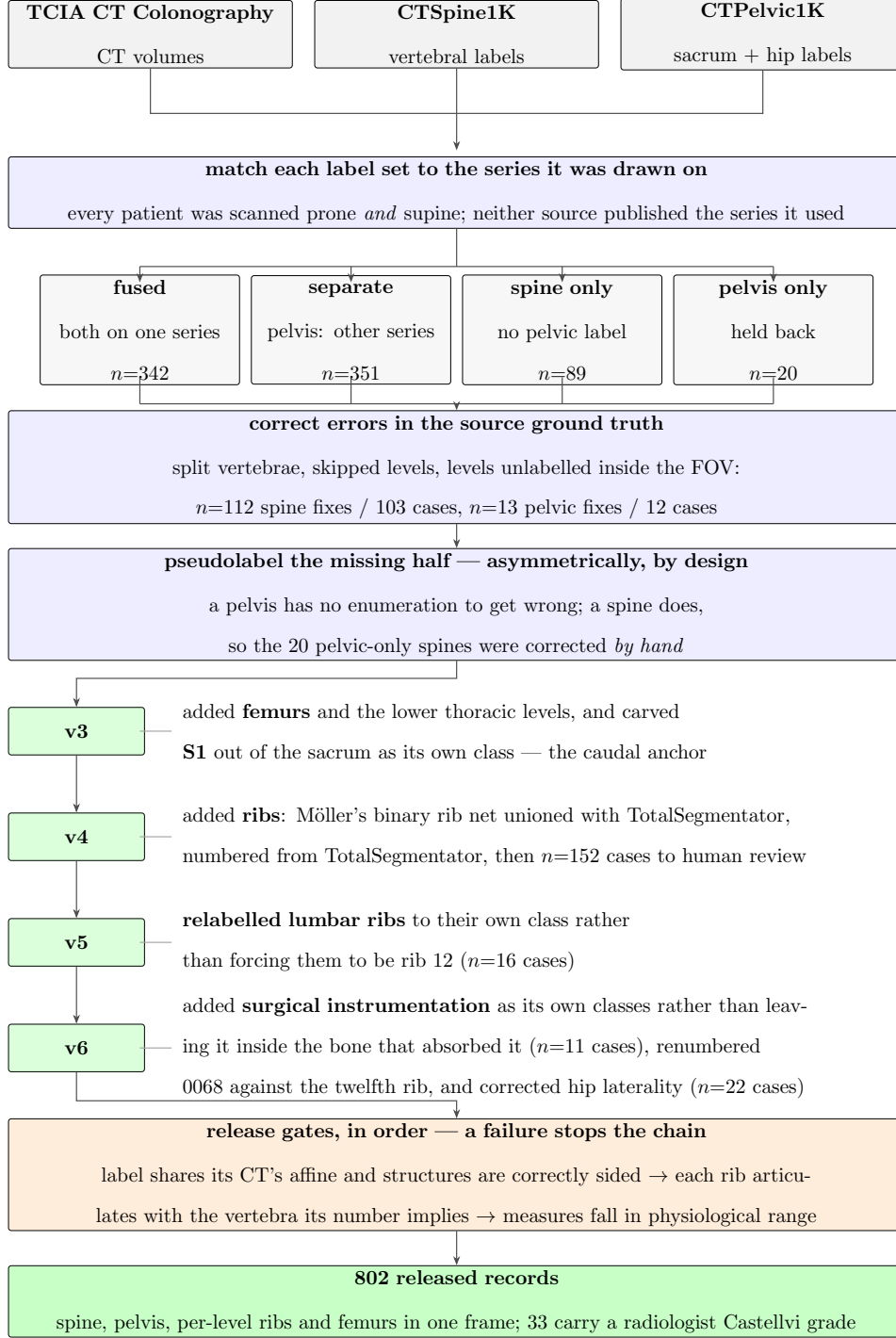


FIG. 1 How the dataset was built. Three public collections are combined patient by patient. Neither annotation source published the CT series its labels were drawn on, so each label set is matched to the series whose bone agrees with it. In 351 of those patients the two collections had drawn their outlines on *different scans of the same person*. Numbers marked n are scans, or corrections, at that step.

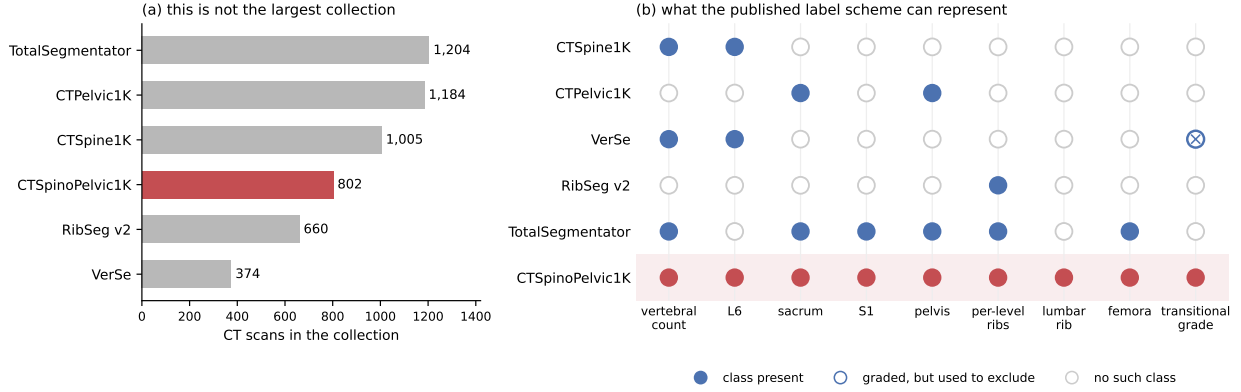


FIG. 2 Public CT collections at the lumbosacral junction. (a) Size, stated first because it is the claim this dataset does *not* make: three of the five comparators carry more scans, and RibSeg v2⁷ labels more ribs. (b) What each published scheme can *represent*, a property of its class list rather than of its segmentation quality. VerSe is graded-but-excluding: it applies the Castellvi classification and uses it to decide what to omit, leaving vertebrae partially fused to the sacrum unsegmented⁵ — 25 of the 33 graded records here.

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

Both CTSpine1K² and CTPelvic1K⁶ draw part of their imaging from the TCIA CT colonography collection^{8,9}, and the vertebral and pelvic annotations in both were produced under board-certified radiologist supervision. Neither released the mapping from an annotation to the specific CT series it was drawn on.

That mapping is not a bookkeeping detail. Every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, often with more than one kernel, so one patient identifier resolves to several volumes differing in posture, position, field of view and noise — and an annotation carries a patient identifier and nothing finer.

Placing it on the wrong volume of the same patient is not merely a displacement. The spine is mildly deformable, and turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so a vertebra moves by a different amount, and in a different direction, from the one above it. Outlines drawn on one series are the wrong *shape* for the other, and no rigid realignment recovers them. In isolation such

82 a mask looks like competent work; it reveals itself only when overlaid on the image, and by
83 then the only remedy is to assign it to the series it was drawn on.

84 The crosswalk therefore proceeds in two stages. Each annotation is first resolved to a
85 patient, and then, within that patient’s volumes only, to the series whose bone *agrees* with
86 it: candidate volumes are thresholded at bone attenuation and scored by overlap with the
87 annotation mask, and the annotation is assigned to the series that maximises it. Bone
88 is used rather than image similarity because bone is what the annotation describes; two
89 acquisitions of the same abdomen resemble each other closely everywhere except in where
90 the skeleton sits.

91 II.B. Fused and split records

92 The two annotation sets were placed independently, so they do not always land on the
93 same series, and the four resulting record types are distinguished because the distinction is
94 the crosswalk’s main finding rather than bookkeeping. Of 802 records, 342 (42.6%) carry
95 spine and pelvic annotations on the same volume and are exported as *fused*. A further
96 351 (43.8%) are *separate*: a pelvic annotation exists for the patient, but it was placed on
97 the other acquisition of the prone/supine pair, so it cannot be combined with the spine
98 annotation without resampling across a change of posture. Only 89 (11.1%) are *spine-only*,
99 meaning no pelvic annotation exists for that patient at all, and 20 (2.5%) are *pelvic-native*,
100 carrying the pelvic annotation alone.

101 The 351 separate records are the direct measurement of the problem described above: in
102 nearly half of all patients the two public collections annotated *different volumes of the same*
103 *person*, and neither collection recorded which. They are retained rather than discarded, and
104 held out of the postural analyses, because a prone and a supine acquisition of one patient
105 differ in spinal alignment and are a resource for a mobility study rather than a defect.

106 II.C. Densification, and why it is asymmetric

107 A corpus in which half the records lack a pelvis is not usable as a spinopelvic dataset, so
108 the missing structures were pseudolabelled. The two directions are not equivalent and were
109 not treated as such.

110 *Pelvis onto spine-only records.* The sacrum and the innominate bones are single objects
111 with no enumeration: there is no count to get wrong, no naming convention to choose,
112 and no analogue of a transitional vertebra. A pelvic segmenter can therefore be trusted in
113 a way a vertebral one cannot, and its quality is reported as held-out performance on the
114 *pelvic_native* records, which were withheld from training for exactly this purpose.

115 *Spine onto pelvic_native records.* The reverse direction carries the whole problem this
116 dataset exists to address. A pseudolabelled spine must commit to a vertebral count, and
117 on a transitional vertebra it will commit to the commoner reading. Those 20 records were
118 therefore not densified automatically: each pseudolabelled spine was reviewed and corrected
119 by hand. Twenty cases is a tractable amount of manual work; four hundred would not have
120 been, which is the practical reason the asymmetry runs the way it does.

121 II.D. Ribs

122 No public rib annotation covers this cohort, and the two available automatic sources fail
123 in complementary ways.

124 Möller’s binary rib network¹⁰ is a detection model of high reported accuracy, but it
125 segments ribs that lie *entirely within* the field of view; a rib clipped by the edge of an
126 abdominal acquisition is liable to be missed altogether. Since this is a colonography cohort,
127 the lower ribs are routinely clipped, and those are precisely the ribs the count depends on.

128 TotalSegmentator⁴ does segment clipped ribs and supplies the per-rib numbering that a
129 binary network cannot. Its characteristic failure is at the other end of the bone: the most
130 medial portion of the rib — roughly the last tenth to sixth approaching the costovertebral
131 joint — is frequently absent. A rib truncated at its medial end never reaches the vertebra
132 it articulates with, so the incidence test that assigns a rib to its level has nothing to test.

133 The two were therefore unioned: TotalSegmentator supplies numbering and the medial-
134 clipped cases, Möller supplies detection where its output is more complete, and the union is
135 carried forward with TotalSegmentator’s per-level identities.

136 The result is a pseudolabel, and the review it received was triaged rather than exhaustive
137 — which is worth stating plainly, because a rib layer described only as “human-corrected”
138 invites the reader to assume every bone was looked at. A label-based rule flagged every
139 record in which one rib bone carried more than one identifier, or in which a rib failed to

reach the vertebra it articulates with; a single bone with two numbers is a numbering error by construction, whatever the anatomy. That rule selected 152 records. 148 were reviewed and corrected by a trained reviewer; 4 were not, and are identified in the release. Reviewers were medical students working through OpenSpineConsortium¹¹, an open-source framework for engaging medical students in computational spine imaging research: annotators are trained against a written labelling protocol, work case-by-case in a shared review tool, and every correction is recorded against the annotator who made it, so the review is auditable rather than anonymous. The remaining records passed the rule and were not individually inspected, so the guarantee the rib layer carries is that it is free of the failure modes the rule detects, and no stronger. The residual rib-vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26), and every non-VerSe structure takes a fixed identifier above that range — S1 carved from the sacrum (29), hips (30–31), femora (32–33), ribs (34–45 left, 46–57 right), soft tissue (58–73), lumbar ribs (74–75) and surgical hardware (76–82).

Two anchors make the count decidable. Rostrally, **T12** is the last rib-bearing vertebra and sits directly above ground-truth L1. Caudally, **S1** is carved as a distinct class, giving the bottom bracket and the sacral endplate landmark required for sacral slope and pelvic incidence. With both brackets present, L5 and L6 are distinguishable where a spine-limited view alone would not be.

How S1 is cut. The sacrum’s outer boundary is the source annotation and is never altered; S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the sacrum’s own cranio-caudal axis — taken as the principal component most closely aligned with the superior–inferior direction, since the bone is nearly as wide as it is tall and its longest axis is therefore not reliably the cranio-caudal one — made orthogonal to the patient’s left–right axis. That left–right axis is measured rather than assumed, as the normal of the reflection plane that best maps the sacrum onto itself, which removes any rotation of the patient within the scanner: across this cohort that rotation has a median of

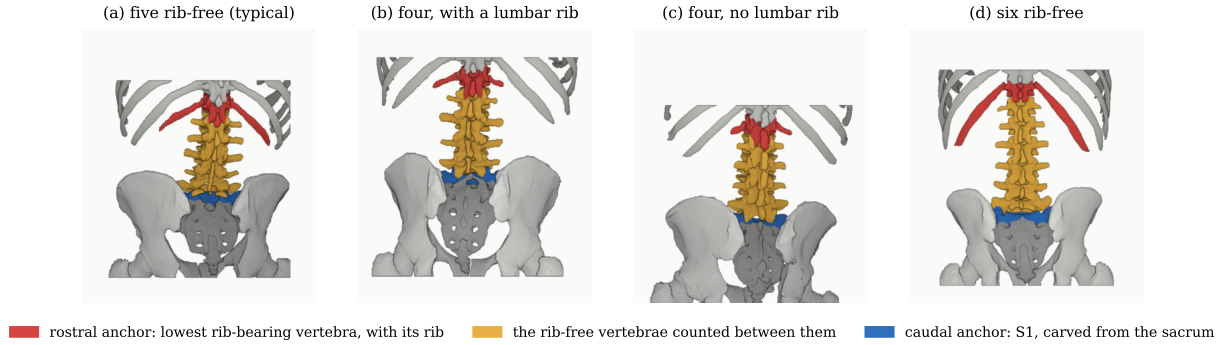


FIG. 3 The two anchors and the interval between them, from the released labels, posterior view. Rostral anchor: the lowest rib-bearing vertebra and the rib that makes it one. Caudal anchor: S1. Both are found from the labels rather than assumed, and panels are coloured by role rather than identifier, since colouring T12 would assert the count the figure derives. Panels (b) and (c) reach the same interval of four from opposite ends and are indistinguishable by it. All panels are at one millimetre scale, aligned at the sacral base.

170 4.5° and reaches 16.1°, and 508 of 801 records exceed 3°. The level of the cut preserves the
 171 sub-division volume of the preceding release, so the plane changes the shape and orientation
 172 of the S1/S2 boundary and not how much of the sacrum is called S1. Per-record geometry
 173 ships with the archive.

174 Neither anchor is novel. TotalSegmentator⁴ carries a separate S1 class and labels ribs per
 175 level, and VerSe³ carries L6 — this release uses the VerSe identifiers, so its L6 *is* VerSe’s,
 176 interoperable with it and with CTSpine1K rather than local to this work. L6 is retained not
 177 because it is new but because a scheme without it cannot record a six-lumbar spine at all,
 178 and must renumber the column or drop a level to fit; TotalSegmentator stops at L5.

179 **A rib on a lumbar body receives its own class**, and this is the one class here with no
 180 published counterpart. A lumbar rib and a stump rib are the same object under two counts,
 181 so assigning it a number would commit the label to one reading of an enumeration anomaly.
 182 A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must
 183 either call it rib 12 — asserting the vertebra beneath it is thoracic, the very question at
 184 issue — or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and
 185 VerSe’s T13 is a vertebra rather than a rib. 16 released records carry a lumbar rib and 18
 186 carry an L6.

Figure 3 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count through a lumbar rib and some without one; the two are indistinguishable by the count alone and are distinguished here only because rib presence is itself labelled.

Sixteen records carry a lumbar rib in the released volumes: thirteen bilateral and three unilateral, all three on the right. That laterality is reported because it is what the release contains and not as a finding — sixteen records cannot support a statement about side preference, and none is made.

Eighteen records carry an L6. Fourteen are labelled lumbarisation, two sacralisation, one semi-sacralisation and one normal, which is worth stating because it shows the two annotation axes disagreeing in the open: a record can carry six lumbar-type vertebrae and still be read as a sacralisation, or as unremarkable, depending on where the reader started the count.

Both counts are derived from the released label volumes, and so are the manifest fields that report them. That is a correction rather than a design note: through v5 the `has_l6` flag was true for exactly one record, which contains no L6, and false for all eighteen that do, while `has_lumbar_rib` was false everywhere and `n_lumbar_labels` read zero in 799 of 802 records. A reader selecting the six-lumbar cases on those fields would have received one record that is not one, and a reader selecting lumbar ribs an empty set, with nothing to indicate either. The fields are rebuilt for this release by counting identifiers 25 and 74–75 in each volume, and `lumbar_rib_side` is added alongside them.

II.F. Computational tools

Access. Every processing step described above is performed by code released under an open-source licence and archived with the dataset; no step depends on software that is not publicly obtainable. The release archive carries the reference loader, the label-scheme definition and the quality-control scripts that generated the tables in this manuscript.

Versions and parameters. Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator⁴ (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller’s¹⁰ released weights, applied through the nnU-Net v2 framework¹². The pelvic completion for records whose pelvis was absent used a

TABLE I Release validation. Gates run in order and a failure stops the chain.

Check	Result
Geometric invariants (all cases)	802/802 pass
label shape, affine, spacing match CT	—
no identifier outside the scheme	—
left/right structures correctly sided	—
Ribs present	11,548
not evaluable (expected vertebra outside the FOV)	5,788
no vertebra within the anchor distance	11
evaluable	5,749
matching the expected vertebra	5,747
residual offset	2 (0.035% of evaluable)
misnumbered cases	0
Spinopelvic medians within published range	6/6

217 five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that
 218 never trained on it, so no record is completed by a model that has seen it. Image handling
 219 throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is
 220 written, so every label shares its image’s grid and affine exactly.

221 *Generative artificial intelligence.* Large-language-model assistance was used for manuscript
 222 preparation — drafting and editing of text, and generation of analysis and figure code —
 223 under author review. It was not used to generate, annotate or interpret any imaging data,
 224 and no scientific claim in this manuscript originates from it. All numbers reported here are
 225 computed by the released code from the released volumes.

226 II.G. Validation

227 Validation is layered, and the ordering matters: measurements computed on a corpus
 228 that has not established internal consistency do not look broken, they look finished.

229 *Geometric invariants.* Every case is checked for label geometry agreeing with its CT in
 230 shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty

label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

It compares *each sided pair separately*, and neither alternative finds what it finds. Ribs alone passed all 802 while four records carried `left_hip` on the patient’s right; pooling the pairs recovered three and missed one, because hips and femora outweigh the ribs by voxel count and a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs present, 5,788 are not evaluable at all — the vertebra their number implies lies outside the field of view and nothing else is within the anchor distance — and a further 11 have no vertebra in reach. That leaves 5,749 with an anchor to check, of which two remain offset (0.035%). The denominator is stated because quoting two against 11,548 halves the rate by counting ribs the check never examined, and because more than half of the ribs in a screening abdominal CT being unevaluable is itself a fact about the corpus worth reporting. Both offsets are field-of-view truncations on individually characterised cases — one at +1 level and one at −1 — and are reported rather than suppressed: a threshold tuned until the count reaches zero yields a cleaner number and a less informative one.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra — not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 74 or 75 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Derived spinopelvic measures are compared against Vialle *et al.*¹³ (Table II). Pelvic incidence is the strongest of these checks because it is a morphological property of the pelvis rather than a posture, and so is directly comparable to a standing reference cohort.

260 II.H. Castellvi typing, and the reading it still requires

261 Every record whose vertebral count departs from five carries a Castellvi grade¹⁴ in the
262 released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. This is
263 released as an annotation layer because the grade and the count are *different axes* and the
264 dataset is built to show that they are: grade IIIb occurs here at rib-free counts of four, five
265 and six alike, and seven of the 33 graded cases carry a perfectly normal count of five. A
266 corpus that recorded only the count would describe those seven as unremarkable.

267 The grading is a single reader’s, with five cases independently read a second time. Three
268 of those five agreed exactly. Of the two that did not, one is Ib against IIb — adjacent
269 subtypes, the ordinary kind of disagreement — and the other is IIIb against IIb, which
270 crosses the boundary between bony fusion and articulation without it. That is the distinction
271 the classification is built on and the one carrying clinical weight¹, so it is reported rather
272 than folded into a single agreement rate. Five second reads establish that the grading was
273 checked; they are far too few to support an inter-rater statistic, and none is quoted.

274 The typing is therefore submitted for confirmation by a board-certified neuroradiologist,
275 blinded to the existing grades and to the vertebral counts, reading all 33 cases. Where
276 the confirming read and the original agree the grade stands; where they disagree the case
277 is adjudicated in a joint reading and the adjudicated grade released, with both original
278 reads retained in the manifest so that no disagreement is silently resolved. Full agreement
279 statistics will be reported for the complete set of 33 rather than for the current subset of five.
280 *This confirmation is in progress and the release is not final until it completes*; the grades
281 described here are the single-reader grades, and the manifest records which reads each grade
282 rests on.

283 II.I. Surgical hardware, and why a threshold is not enough

284 Metal is trivial to detect and hard to interpret, and the difference matters here for one rea-
285 son: **an iatrogenic fusion is indistinguishable from a congenital one to a distance**
286 **measurement**. A cage-bridged interspace reads as “no gap” exactly as a congenitally fused
287 transitional vertebra does, so an unrecorded instrumented case contributes a false positive to
288 the central measure — and until this release nothing said which cases they were: identifiers

TABLE II Derived measures against a published asymptomatic reference ($n = 260$, standing)¹³. This cohort is supine, and the two postural measures depart in opposite directions — sacral slope 4.1° lower, pelvic tilt 4.3° higher. That is one discrepancy, not two: $PI = PT + SS$ holds by construction, so with pelvic incidence at its published value any fall in one appears as an equal rise in the other.

Measure	This dataset (supine)	Reference (standing)
Pelvic incidence	54.7°	54.7 ± 10.6
Pelvic tilt	17.3°	13 ± 6
Sacral slope	36.9°	41.0 ± 8.4
Lumbar lordosis (supine)	52.7°	43 ± 11.2 (standing)
PI – LL mismatch	2.6°	centred near 0

76–79 were declared in every prior version and populated in none.

II.I.0.a. Detection over-calls by a factor of seven. A threshold at 1800 HU inside a shell around the labelled skeleton flags 84 of the 802 records. Raising it to 2500 HU — the lower of the two values validated in the metal-segmentation literature — leaves 52 with a component above a 40-voxel floor. A radiologist read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Instrumentation prevalence in this cohort is 1.4%, not the 10.5% an unreviewed threshold reports. The threshold at 3000 HU is not the answer either: these series clip at 3071 HU, so it measures the saturation plateau rather than the implant.

II.I.0.b. Saturation does not separate implant from artefact. Both groups reach the scanner ceiling and 4 rejected proposals exceed it, peaking at 11,798 HU — the signature of reconstruction overshoot rather than of denser metal. What separates them is position and size together: a real surgical site *and* volume, every confirmed implant at $2,586 \text{ mm}^3$ or above and every rejection at $1,768 \text{ mm}^3$ or below (Fig. 4).

II.I.0.c. The subtype block had to be extended. Identifiers 76–79 name spinal instrumentation — generic, cage, screw-and-rod, plate — and cannot describe what this cohort holds, so **80 arthroplasty**, **81 sacroiliac screw** and **82 osteosynthesis** were added. The distinction is clinical, not cosmetic: osteosynthesis holds parts of the same bone together where arthroplasty replaces a joint, and without it the shape rules call a femoral stem a rod,

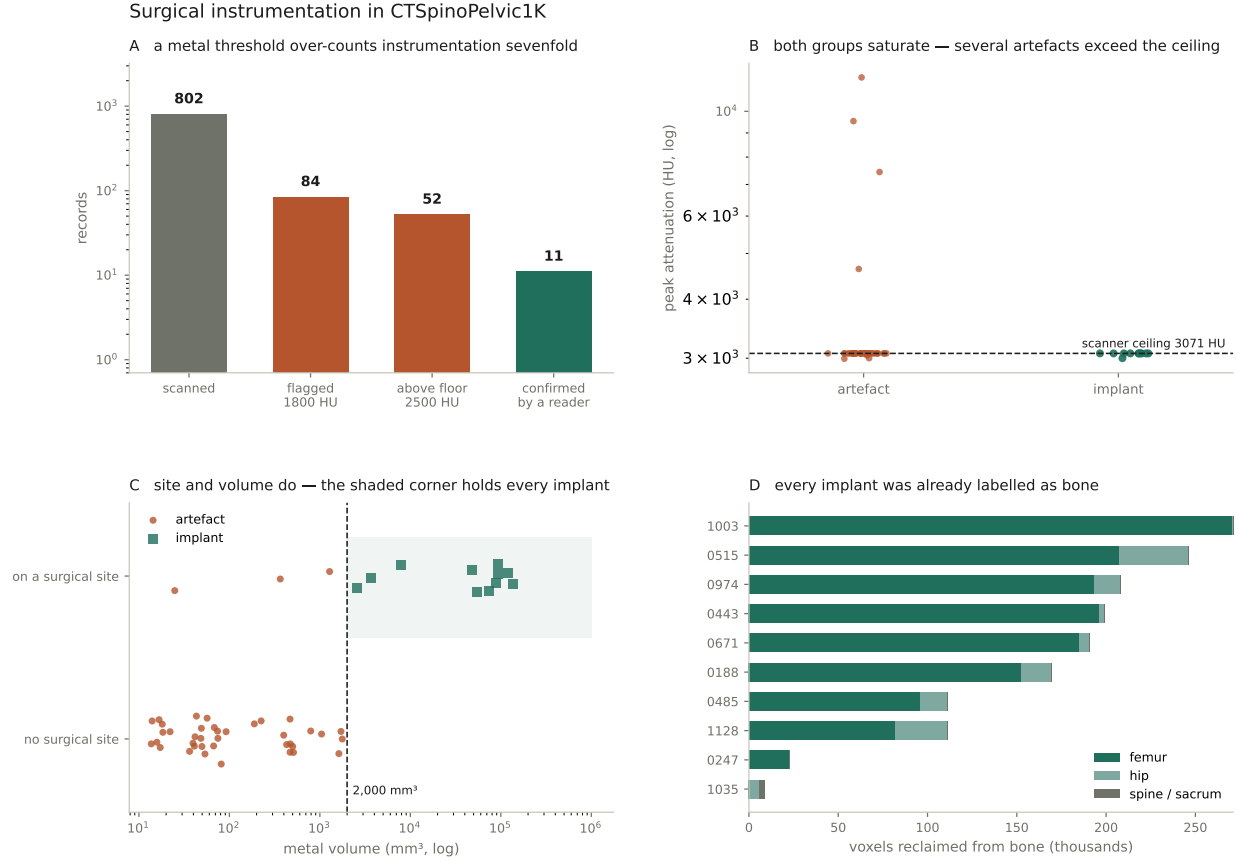


FIG. 4 Surgical instrumentation. (a) An unreviewed metal threshold over-counts instrumentation sevenfold. (b) Saturation cannot sort implant from artefact: both reach the scanner ceiling and several artefacts exceed it. (c) Site and volume can. (d) Every confirmed implant was already labelled as bone, so naming it reclaims voxels from the structure that held them.

since a stem is long and thin and that is all “linear” tests. The confirmed cases comprise 8 arthroplasties, one osteosynthesis (three parallel cannulated screws in an inverted-triangle configuration at the femoral neck), one sacroiliac screw fixation and one pair of threaded cylindrical interbody cages.

II.I.0.d. Metal outranks bone, and that is a subtraction. In every confirmed case the implant was already inside a bone label, a segmenter handed a bright object against a cortical surface having taken it for bone. Naming the hardware therefore reclaims voxels rather than adding them: 1,538,852 voxels across the eleven records. The consequence reaches past bookkeeping. Pelvic incidence and pelvic tilt are measured from the femoral head centre, and in 8 of these cases that head is an implant — any spinopelvic parameter

318 previously computed from them was measured on metal.

319 III. DATA FORMAT AND USAGE NOTES

320 Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canon-
321 icalised to PIR, requiring no resampling. Masks are NIfTI rather than DICOM: the policy
322 prefers DICOM where applicable, and for volumetric segmentation masks NIfTI is what the
323 downstream tooling expects.

324 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the
325 data as `splits_5fold.json`. The five validation folds partition all 802 records — they are
326 disjoint and their union is the cohort — so the release provides cross-validation and *not* a
327 held-out test set. Anyone reporting a single held-out number should carve one out and say
328 which records it holds, because the folds shipped here will not do that job.

329 Stratification is on the transitional subtype rather than on a binary LSTV flag, and it
330 enforces a minimum of each rare subtype in every validation fold: each fold’s validation
331 set carries three sacralisation and three lumbarisation records, so no fold is evaluated on a
332 cohort missing the class the dataset exists for. With 15 lumbarisation and 15 sacralisation
333 records across 802, that is the practical limit of what stratification can guarantee, and it
334 is the reason a fold-level metric on the rare classes has an error bar far wider than its own
335 decimal places.

336 Table III lists the key data types and metadata against the fraction of records for which
337 each is populated. Two entries are worth reading carefully. The Castellvi grade is present
338 for 4.1% of records because it is defined only for the transitional ones; among those it is
339 complete. A pelvic series identifier is present for 88.9% because the crosswalk records it only
340 where a pelvic annotation was actually placed, and a user joining on it should expect the
341 shortfall rather than treat it as loss.

342 The archive of record is Zenodo (10.5281/zenodo.22139642). Code and documentation
343 are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror,
344 not the archive.

TABLE III Key data types and metadata, and the fraction of the 802 released records for which each is populated. Counts are obtained by reading the released manifest, not by assertion. A field below 100% is not an error: age and sex are absent where the source collection did not publish them, a pelvic series identifier exists only where a pelvic annotation was placed, and a Castellvi grade exists only for the transitional records that were graded.

Field	Records	Available
Identifier, image path, label path, configuration, match type	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class	802	100.0%
Spine and pelvic annotation origin	802	100.0%
Instrumentation and partial-annotation flags	802	100.0%
Image/label alignment check	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade	33	4.1%

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value for patient sex, which is a recorded value and not a missing one, and 53 carry no sex field at all. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals, and records outside the female/male dichotomy are excluded from measures stratified by sex — a limitation of those measures rather than a property of the cohort, and stated here because folding eleven people into “unrecorded” would misdescribe what the source says. The lower age bound is the screening guideline rather than a selection applied here, and it is why the measures below sit closer to a reference range than a surgical series would.

Count-free measures (Fig. 5). Five rib-free vertebrae above the sacrum is typical; four

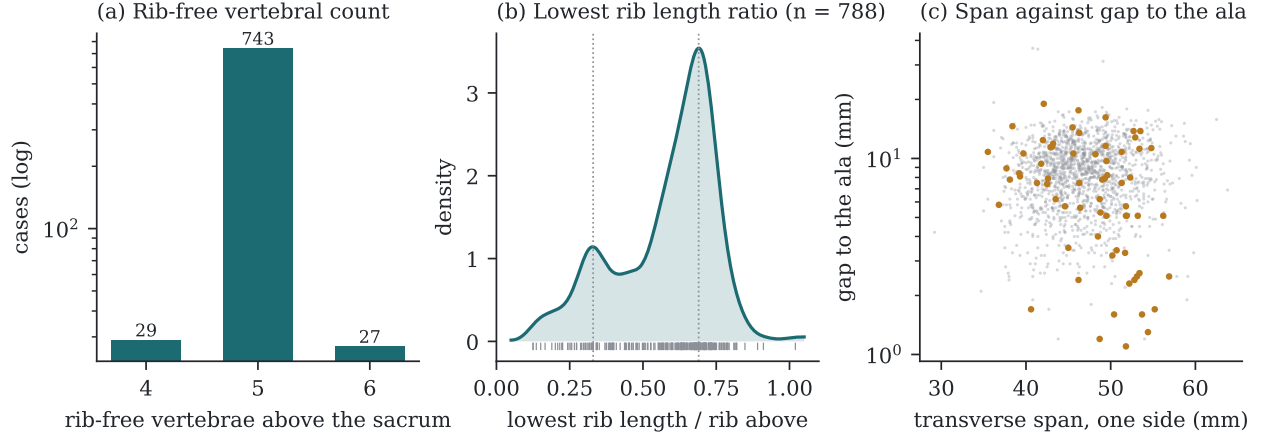


FIG. 5 Count-free measures, valid without knowing which vertebra is which. (a) Vertebrae between the lowest rib and the sacrum: an interval, not a level assignment. (b) Lowest rib length as a fraction of the rib above, showing two populations near 0.33 and 0.69 rather than one distribution with a tail. (c) Lowest lumbar transverse span against its gap to the sacral ala; cases carrying a source transitional label are marked.

and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, with modes near 0.33 and 0.69.

Level gradients and spinopelvic measures (Fig. 6). Vertebral dimensions increase caudally in step with the load each level carries. Pelvic incidence does not vary with age, while pelvic tilt increases — the descriptive signature of a pelvis retroverting as a spine ages.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.¹⁵ L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*: spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, which makes sacral morphology measurable against the lumbar column rather than in isolation — the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible

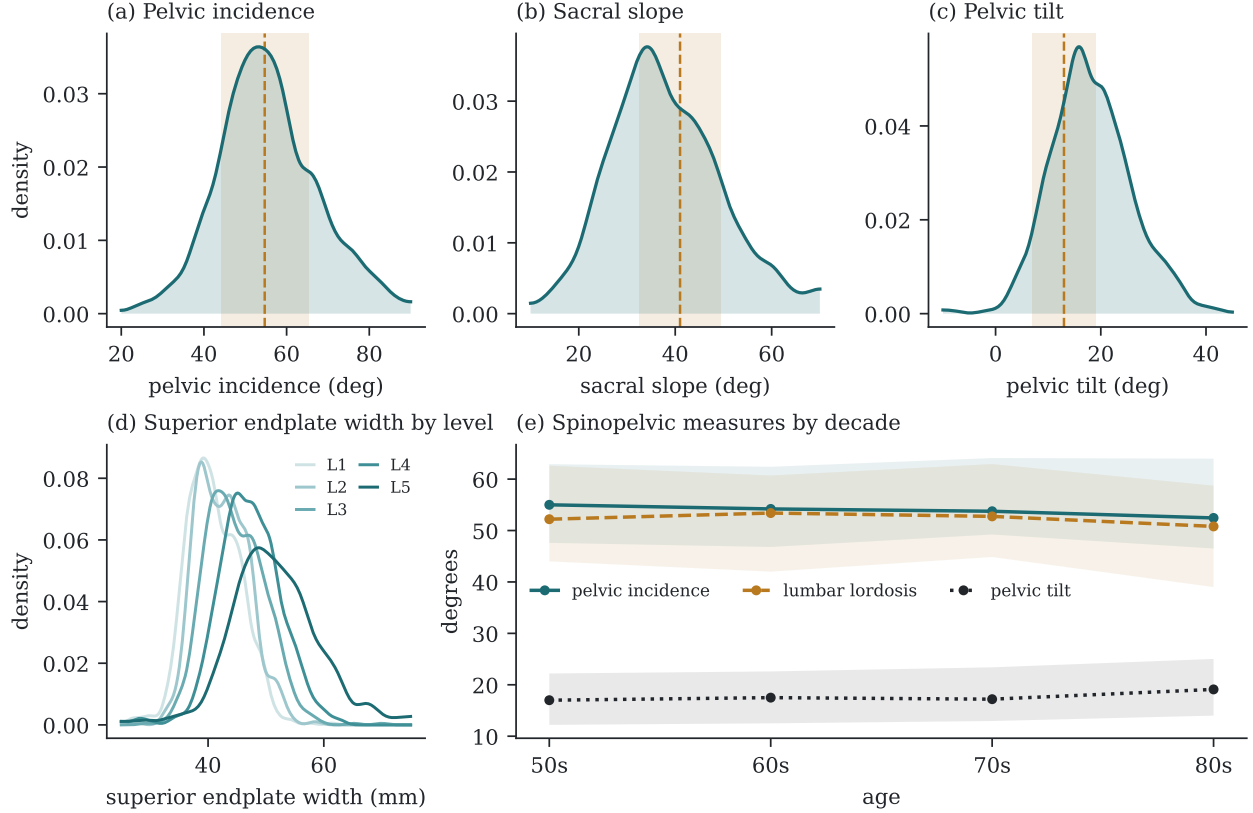


FIG. 6 (a–c) Spinopelvic measures against published reference bands (mean $\pm$ SD). (d) Superior endplate width by level. (e) Median and interquartile range by decade; pelvic incidence is flat by construction of the anatomy, which makes it a negative control for the measures beside it.

from the labels themselves.

The transitional layer serves that use rather than competing with it. A corpus assembled purely for enumeration anomalies would want whole-spine coverage this cohort lacks; what these records support is planning *across* a junction, and no trajectory can be planned across a junction whose segments cannot be named. Level-numbering benchmarks, variant studies and opportunistic screening follow from the same annotation.

V.A. Three uses the released measurements support

Naming a level from its own geometry. Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this cohort was assembled around.¹⁶ The failure is structural rather than careless: the count is ambiguous exactly where the variant sits. Per-level body height, canal width, endplate

width, transverse-process span and wedge ratio are released for T11–L5 across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained on them is not learning the convention that fails.

Reference morphometry at cohort scale. Reference values for vertebral and canal dimensions derive substantially from cadaveric series and small imaging cohorts;¹⁷ these are computed identically across 802 records from segmentations. Larger n improves the precision of a distribution without fixing the population it came from, and this one is a screening cohort aged 50 and over.

Patient-specific models. Planning is moving toward patient-specific biomechanical models rather than population norms.¹⁸ Such a model needs spine, pelvis and femoral heads in one frame, which is this release’s construction; in 351 patients the annotation sits on two acquisitions in different positions, giving a within-patient change in alignment against which a predicted postural response can be checked.

Its limitations are part of it, and they are stated at the level of detail a reader would need to catch them independently.

Coverage. Thoracic ground truth is field-of-view limited and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

Declared but empty, and confirmed so. The soft-tissue identifiers (58–73) are declared in the scheme and occur in no record — checked by counting identifiers in all 802 released volumes rather than by assertion. They are retained so that a downstream annotation can use them without renumbering, and a user should treat their absence as absence of annotation rather than absence of the structure. The hardware identifiers were in this position until v6 and are no longer: 76–82 are populated in the eleven instrumented records described in Sec. II.I.

Strength of the labels varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabelled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated — which is

precisely why the measures reported here are count-free — and the Castellvi grades are a single reader’s pending the confirmation described in Sec. II.G.

Structures are not universally present. The same census shows the release is not uniformly complete. Sacrum, hips and femora are present in all 802 records. One record carries no S1, one lacks T12, and nine lack an L5 identifier — in every case because the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labelled L6 or incorporated into the sacrum. A user filtering on the presence of the caudal anchor should filter explicitly rather than assume it.

The partial-annotation sentinel is unused. Identifier 255 marks a region that was not traced and is not background. In the preceding release exactly one record used it; that record has since been completed and no record in this release carries it. The identifier is retained because the protocol is part of the scheme and a later release may again ship a partially traced record.

Splits. The shipped folds are cross-validation folds covering the whole cohort. There is no held-out test set, and with 15 lumbarisation and 15 sacralisation records a per-fold metric on the rare classes is estimated from three cases.

VI. DISCUSSION

What the release enables. Three kinds of reuse need no further annotation: segmentation and level-numbering benchmarks on a cohort where the count is genuinely ambiguous rather than merely unlabelled; morphometric description of the lumbosacral junction from landmarks inside the field of view; and templating work on the instrumented subset, where metal is labelled rather than absorbed into the bone beside it. Because spine, pelvis, ribs and femora share one coordinate frame, a measurement needing both a spinal and a pelvic landmark — sacral slope and pelvic incidence among them — requires no registration between two annotations.

Limitations with respect to future use. The cohort is a colon-screening population imaged supine and prone — not a trauma, deformity or paediatric series — so generalisation beyond adult abdominal computed tomography is not established here. No scan contains C2, so the conventional cervical-down count cannot be performed on any record; that is a property of the imaging rather than the annotation, and it is why the measures reported here are count-

free. Label strength is not uniform: vertebral labels derive from radiologist-supervised sources, pelvic labels on records that lacked one are model-completed, and the rib layer is a pseudolabel whose review was triaged rather than exhaustive. The sacral sub-division inherits its level from an automatic method, and in 100 records that level places S1 above half the sacrum or below fifteen percent of it, which no first sacral segment is; those records are flagged in the released quality-control table and should be filtered before any measurement depending on the sacral endplate.

Comparison with related material. Sources annotating the spine alone cannot supply a pelvic landmark, and sources annotating the pelvis alone cannot supply a vertebral count. The contribution here is the crosswalk placing both on the same series, with classes for the anatomy an enumeration anomaly actually produces: where another scheme must record a sixth lumbar vertebra or a lumbar-borne rib as something else, this release records it as itself.

VII. CONCLUSION

CTSpinoPelvic1K is an openly licensed release of 802 abdominal computed-tomography records placing spine, pelvis, per-level ribs and femora in one coordinate frame, in which the anatomy that makes lumbar numbering ambiguous — a sixth lumbar vertebra, a lumbar-borne rib, a transitional lumbosacral junction — carries a class of its own rather than being recorded as something it is not. Two counting anchors are explicit, the last rib-bearing thoracic vertebra above and a sacral sub-division below, so a level can be identified from a field of view that cannot support the conventional count. The release is archived under a persistent identifier, ships the loader and quality-control tables needed to verify it, and states per structure how strong each label is.

DATA AVAILABILITY

The dataset is permanently archived at <https://doi.org/10.5281/zenodo.22139642>, which resolves to the current version; the release described here is v7 (10.5281/zenodo.22242745). The archive carries the reference loader, the label-scheme definition and the quality-control tables, and the analysis code is distributed with it under an open-source

472 licence. The repository URL is omitted here because it identifies the authors; it is given on
473 the title page and will be restored to this section for the camera-ready version.

474 CONFLICT OF INTEREST

475 The authors have no relevant conflicts of interest to disclose.

476 ETHICS

477 This work uses publicly available, de-identified imaging from The Cancer Imaging
478 Archive^{8,9} and annotations derived from it. The investigators obtain no identifiable private
479 information and have no interaction with the individuals imaged, so the activity does not
480 meet the definition of human participant research under 45 CFR 46.102. A determination
481 to that effect has been submitted to the authors' institutional Review Board.

482 REFERENCES

- 483 ¹G. P. Konin and D. M. Walz, Lumbosacral transitional vertebrae: classification, imag-
484 ing findings, and clinical relevance, *AJNR Am. J. Neuroradiol.* **31**, 1778–1786 (2010),
485 doi:10.3174/ajnr.A2036.
- 486 ²Y. Deng et al., CTSpine1K: a large-scale dataset for spinal vertebrae segmenta-
487 tion in computed tomography, *Mach. Learn. Biomed. Imaging* **3**, 824–832 (2025),
488 doi:10.59275/j.melba.2025-gf84.
- 489 ³A. Sekuboyina et al., VerSe: a vertebrae labelling and segmentation bench-
490 mark for multi-detector CT images, *Med. Image Anal.* **73**, 102166 (2021),
491 doi:10.1016/j.media.2021.102166.
- 492 ⁴J. Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomic structures
493 in CT images, *Radiol. Artif. Intell.* **5**, e230024 (2023), doi:10.1148/ryai.230024.
- 494 ⁵H. Liebl et al., A computed tomography vertebral segmentation dataset with anatomical
495 variations and multi-vendor scanner data, *Sci. Data* **8**, 284 (2021), doi:10.1038/s41597-
496 021-01060-0.

- ⁶P. Liu et al., Deep learning to segment pelvic bones: large-scale CT datasets and baseline models, *Int. J. Comput. Assist. Radiol. Surg.* **16**, 749–756 (2021), doi:10.1007/s11548-021-02363-8.
- ⁷L. Jin et al., RibSeg v2: a large-scale benchmark for rib labeling and anatomical centerline extraction, *IEEE Trans. Med. Imaging* **43**, 570–581 (2024), doi:10.1109/TMI.2023.3313627.
- ⁸K. Smith et al., Data from CT COLONOGRAPHY, The Cancer Imaging Archive, 2015, doi:10.7937/K9/TCIA.2015.NWTESAY1.
- ⁹K. Clark et al., The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository, *J. Digit. Imaging* **26**, 1045–1057 (2013), doi:10.1007/s10278-013-9622-7.
- ¹⁰H. Möller et al., Automated thoracolumbar stump rib detection and analysis in a large CT cohort, 2025, arXiv:2505.05004. doi:10.48550/arXiv.2505.05004.
- ¹¹A. Schehr, J. Kim, and G. Schwing, OpenSpineConsortium: an open-source framework for medical student engagement in computational spine imaging research, *Cureus* (2026), Published 14 July 2026. doi:10.7759/cureus.112661.
- ¹²F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation, *Nat. Methods* **18**, 203–211 (2021), doi:10.1038/s41592-020-01008-z.
- ¹³R. Vialle, N. Levassor, L. Rillardon, A. Templier, W. Skalli, and P. Guigui, Radiographic analysis of the sagittal alignment and balance of the spine in asymptomatic subjects, *J. Bone Joint Surg. Am.* **87**, 260–267 (2005), doi:10.2106/JBJS.D.02043.
- ¹⁴A. E. Castellvi, L. A. Goldstein, and D. P. K. Chan, Lumbosacral transitional vertebrae and their relationship with lumbar extradural defects, *Spine (Phila. Pa. 1976)* **9**, 493–495 (1984), doi:10.1097/00007632-198407000-00014.
- ¹⁵P. J. Pickhardt, B. D. Pooler, T. Lauder, A. M. del Rio, R. J. Bruce, and N. Binkley, Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications, *Ann. Intern. Med.* **158**, 588–595 (2013), doi:10.7326/0003-4819-158-8-201304160-00003.
- ¹⁶N. E. Epstein, A perspective on wrong level, wrong side, and wrong site spine surgery, *Surg. Neurol. Int.* **12**, 286 (2021), doi:10.25259/SNI.402_2021.
- ¹⁷E. C. Benzol, *Biomechanics of Spine Stabilization*, Thieme, New York, 3rd edition, 2015.

529 ¹⁸Digital twins: a “tremendous step forward” for complex spine cases, Becker’s Spine
530 Review, 2026, [date] 2026. Quoting D. Kojo Hamilton, MD, University of Pittsburgh
531 School of Medicine. Accessed 2 September 2026. [https://www.beckersspine.com/spine/
532 digital-twins-a-tremendous-step-forward-for-complex-spine-cases/](https://www.beckersspine.com/spine/digital-twins-a-tremendous-step-forward-for-complex-spine-cases/).